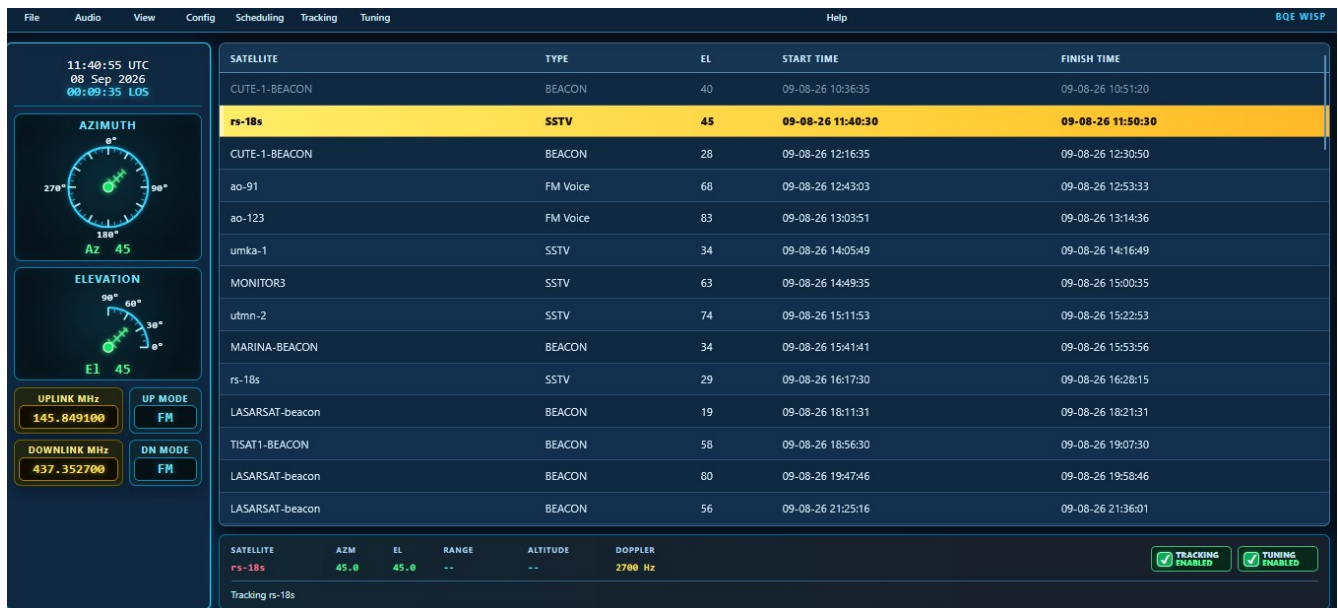


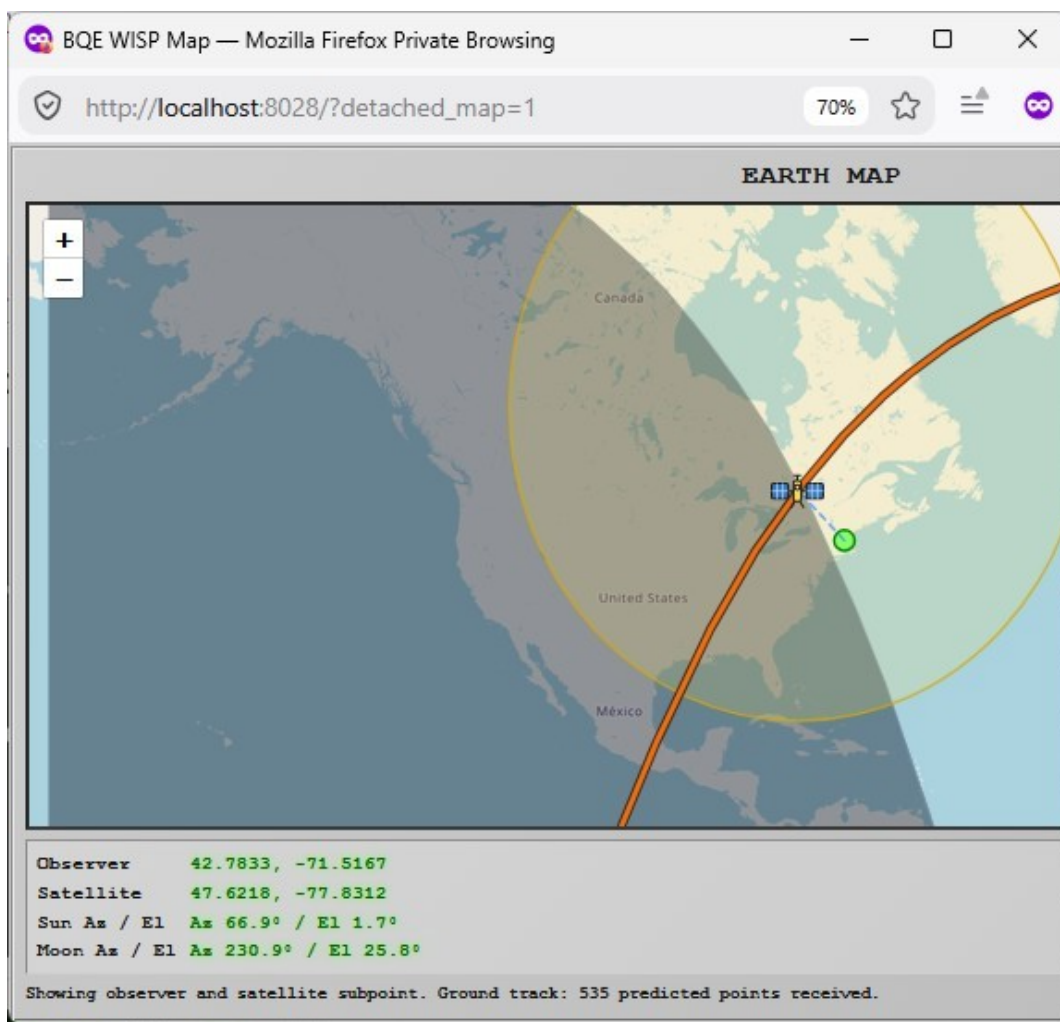
BQE Wisp

LEO Satellite Scheduling, Tracking and Tuning software for Linux and Windows.



BQE Wisp is a Sunday afternoon automation project that got out of control, and over time grew into a from scratch re-implementation of the major functionality of Chris Jackson's Wisp32, which was the Gold Standard for satellite tracking throughout the 1990's, but which is no longer maintained.

In its current form, BQE Wisp is a collection of scripts and data files that run on both Windows and Linux. Both a command line interface, and web UI with a basic tracking map are available.



Other major features include the ability to record passes and decode CW beacons. An experimental version of “Full Doppler Tuning” is available in this version as well.

Acknowledgements

- My longstanding Gratitude and admiration to Chris Jackson, ZL2TPO for his original version all those years ago, which inspired me to pursue Satellites and Automation.
- Also, a major Thank You to Tony Thompson – G1RBO who has spent a significant effort to test BQE Wisp, as well as for his collaboration to support his own G1RBO radio control programs.

License

As the original Wisp software appears to be abandoned, I decided to write my own, and to make it freely available, with the stipulation that it not be used, modified or sold for commercial purposes. For this reason, I cannot call it “Open Source”, as true Open Source would apply to commercial activities as well. That said, I have attempted to be as permissive as possible for non-commercial end educational use. The exact license is included in the program folder as License.txt, as well as via the web UI as Help → License.

Other provisions:

- If you find and fix a bug, or modify the software, please share the updates back to me.
- If you like the software and are using it, please drop me an email, or a digital qsl card – I like to know that folks are enjoying it.
- Please consider making a donation to the Amsat organization of your choice.

High level description of operation

BQE Wisp allows a user to define one or more low orbit Amateur Radio satellites in terms of their favorite frequencies and operating modes. These definitions are known as “nicknames”. BQE Wisp will match these nicknames to orbital tracking data, and generate a schedule of times when the satellite(s) are visible in your location.

As each satellite comes into view, the users radio will be automatically tuned to the uplink and downlink frequencies, which will then be automatically updated to account for Doppler shift.

Between passes, BQE Wisp can optionally be configured with an “Idle Task” that will tune to the radio to a specified “preset” (i.e. a software memory channel) allowing you to do other things, such as monitor a repeater, or other frequencies of interest. BQE Wisp allows presets, and the “Idle Task” to

run a program such as mmsstv (qsstv for Linux), or wsjtx etc. to allow for a wide variety of automated activities between passes.

Nicknames (Important Concept)

Orbital Tracking data (Aka Keps), as downloaded from the internet, consist of a mathematical representation of a satellites orbit. Keps do not, however, contain any frequency or mode information. **BQE_WISP uses the notion of a “nickname” to tie a specific satellite operating configuration to the orbital tracking data.** This is done in a file called bqe_config/satellites.yaml. This file can be updated by the user as desired. Multiple operating modes may also be specified for satellites which switch modes (ISS). The only requirement is that nicknames be unique, and not contain whitespace or strange characters.

Presets are non-satellite “channel configurations” which can be called between passes to perform other automated activities. Channels also use the concept of a nickname to refer to the preset channels.

Presets are stored in a subfolder called presets. The naming convention is **<nickname>_preset.yaml**, where the nickname matches the nickname listed inside the file.

Note that the existing presets are based on my personal preferences, and assume U.S. frequency allocations. A user is free to define any presets they desire, subject to their countries frequency allocations, and the capabilities of the target radio. Unwanted presets can be and can be renived by removing the files from the presets folder, and restarting the program.

Supported Platforms

Windows 10 & 11, as well as recent versions of Linux (Ubuntu etc.)

System Requirements

Recent Windows / Linux PC with Python3 (> 3.11), and Hamlib installed and visible within the users path.

Some python libraries may need to be installed, if they are not already present. The required libraries can generally be installed via the following pip command executed from a Windows Command Prompt window, or linux shell.

python -m pip install PyYAML numpy skyfield requests soundcard lameenc

How to get started

Initial configuration

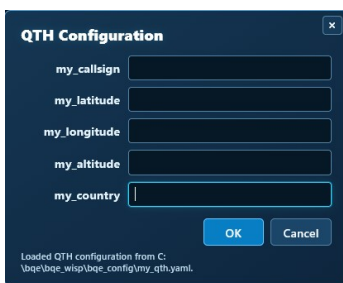
Note: Earlier versions of BQE Wisp required initial yaml file configuration (Rig, QTH etc.) to be done outside of the program. The newer versions generally will allow the server to start, and allow you to make the configuration changes from inside the program. In the event that this does not work, Instructions for manually editing the configuration files are shown below in the Command Line section.

Getting Started in Browser mode (recommended)

- Unzip the code into a folder (i.e. `c:\bqe\bqe_wisp` etc.)
- Open a command window and cd into the folder. (`cd \bqe\bqe_wisp`)
- Type **python bqe_wisp.py** You should see messages indicating that the server has started as well as other informational messages.
 - If the server does not start, you may need to install additional Python libraries (See pip command above) or edit some of the configuration files manually outside of the program. I'm happy to help in these cases.

Open a browser and navigate to <http://localhost:8028> and perform the following steps

- Select the **Config** menu bar item, and update and save both the Radio and QTH information. This information will be written out to `bqe_config\my_rig.yaml` and `bqe_config\my_qth.yaml`



The screenshot shows a 'QTH Configuration' dialog box with a dark blue background. It contains five text input fields labeled 'my_callsign', 'my_latitude', 'my_longitude', 'my_altitude', and 'my_country'. At the bottom right are 'OK' and 'Cancel' buttons. A small text note at the bottom left states: 'Loaded QTH configuration from C:\bqe\bqe_wisp\bqe_config\my_qth.yaml.'



The screenshot shows a 'Radio Configuration' dialog box with a dark blue background. It contains five text input fields labeled 'radio_type', 'radio_port', 'radio_baud', 'radio_ft736r_intermediate_freq', and 'radio_idle_preset_nickname'. The 'radio_type' field has a dropdown menu showing '1'. The 'radio_port' field shows 'com3', 'radio_baud' shows '9600', 'radio_ft736r_intermediate_freq' shows '50.1', and 'radio_idle_preset_nickname' shows '2m_ssb'. At the bottom right are 'OK' and 'Cancel' buttons. A small text note at the bottom left states: 'Loaded Radio configuration template from C:\bqe\bqe_wisp\bqe_config\my_rig.yaml.'

- After this has been updated, restart the server, either by using **File->Restart Server** in the browser menu bar, or hitting ctrl-c in the command window and re-running the **python bqe_wisp.py** command.
- If the server was shutdown without the restart option, return to the command window and restart it by again typing **python bqe_wisp.py**
- Enter the Web console again and execute the **Tracking->Update Keys** menu item, followed by Tracking/ Autoschedule passes. (Update keys could take up to 10 seconds. Autoscheduling passes could take a full minute depending on the speed of your computer.)
 - Satellite schedules often overlap. In the event of conflicting passes, the satellite with the highest pass maximum pass elevation will be selected. (At some future point, I hope to implement a more advanced priority system.)

General operation

Note: FT-736r users should make sure the satellite switch is set to “off” on the radio before running to allow for proper idle time switching to non-satellite modes between passes.

Once you have performed the configuration steps above, starting the server using **python bqe_wisp.py** should bring up the server. The server will read the schedule and automatically track and tune passes. This will happen regardless of whether a browser session is currently active. ***Note that the command window needs to stay open while the server is running.*** You may close or refresh browser sessions as desired. As long as the background python server is running, scheduled passes will be run.

As each pass progresses, tracking and tuning data is calculated every 30 seconds when the satellite is at low elevation, as Doppler and position do not change much. During times when the satellite rises above 65 degrees of elevation, the program will automatically switch to perform tracking and tuning actions every 15 seconds to better keep up with the changing frequency and position, while attempting to minimize start/stop wear on any az/el rotator systems.

The program can be terminated at any time by hitting ctrl-c, and can be restarted cleanly using the existing schedule, even if some passes are already in the past.

When a pass, or idle time interval ends, any companion program started by bqe_wisp will be stopped as well.

Additional notes

The Original WISP supported the Kansas City Tracker (An ISA card from the 1990's) as well as supplying tracking info by Windows DDE. Both technologies appear to be totally deprecated at this point.

The Antenna tracking supplied with this version of bq-e-wisp is currently tied to my personal switched array (i.e. No moving parts) antenna controller, and is likely not of use to anyone other than myself. I will be moving this functionality out of BQE Wisp, and replacing it with hamlib rotctld support in the next couple of weeks.

Presets:

As described earlier, presets are used to define configurations for the radio to use when not actively tracking satellites. Both at program startup, and after each path, the idle task will tune to the preset name listed in `my_qth.yaml`. Other presets can be dynamically selected via the "View → Show Presets" menu bar item, which will expose a panel containing a list of configured presets. These can be used as memory channels to tune around while waiting, if a temporary alternative to the idle task is desired. After the next pass, the idle task preset will again be run. Preset display is off by default, and must be enabled from the menu bar when desired.



SSTV Gallery

BQE Wisp has a simple mechanism to display downloaded SSTV images from a storage area defined by the user. Typically this will correspond to the download area from the SSTV helper program being used. (For example `c:\ham\mmsstv\history`) This can be changed in **general_settings.yaml**.

```
! general_settings.yaml X
C: > bqe > bqe_wisp > bqe_config > ! general_settings.yaml
1  program_settings:
2
3    - common:
4      | python_path: python # Can be a full path
5
6    - bqe_wisp:
7      | horizon_threshold_elevation: 0.1 # degrees
8      | ui_refresh_interval: 5 seconds
9      | ui_port: 8028
10     | sleep_interval: 30 # Seconds
11     | sleep_interval_high_elevation: 10 # Seconds
12     | sstv_gallery_location: 'c:/ham/mmsstv/history' # Full or relative path
13
```

New Features

New User Interface The Ui has been modernized as of this version, but the classic UI is still available by changing the **ui_theme** setting in `bqe_config/general_settings.yaml`. There is no difference in functionality between the 2 user interfaces.

```
- bqe_wisp:
  | horizon_threshold_elevation: 0.1 # degrees
  | ui_port: 8028
  | ui_refresh_interval: 5 seconds
  | ui_theme: modern # Use "modern" or "classic"
```

Pass Recording Passes may be recorded on a per satellite basis by adding the following lines to the appropriate satellite entries in `satellites.yaml`.

```
program_to_run_during_pass:
- python.exe
- plugins/bqe_sound_recorder/bqe_sound_recorder.py
- --input-device
- USB Audio CODEC
- --output
- plugins/bqe_sound_recorder/recordings/<sat name prefix>
```

The **--input-device** is the name the sound card presents to the operating system. “**USB Audio Codec**” is the name presented by current generation Yaesu Rigs. (i.e. FT-991A) and can be seen in the windows device manager.

This will save a timestamped recording of the sound card in mp3 format in the **plugins/bqe_sound_recorder/recordings** folder. Note that recordings will continue to accumulate in this folder, and must be cleaned up manually if desired.

Recordings can be accessed via the “View => Recordings” menu item.

BQE WISP Recordings — Mozilla Firefox

Not Secure http://192.168.1.32:8028/recordings

70% ☆ ☰

Recordings

Folder: C:\bqe\bqe_wisp\plugins\bqe_sound_recorder\recordings

Decode Checked Morse Sort by Filename Sort by Timestamp Refresh

Now playing: iss_2026-09-08_06-58-16.mp3

■ Digital white-noise reduction Off — original audio

Signal Activity Timeline

Detection sensitivity

31 likely signal regions; 0:37 highlighted. Click the timeline to seek.

0:00 10:14

Likely white noise Likely signal activity Playback position

Live Oscilloscope

Morse Decoder Check one or more recordings below, then select Decode Checked Morse. The decoder follows the strongest keyed tone as its pitch changes with Doppler.

Morse	Recording	Last modified	Size	Morse decode
☒	iss_2026-09-08_06-58-16.mp3	2026-09-08 07:09:12 Coordinated Universal Time	15.0 MB	Not analyzed
☐	iss_2026-09-08_05-21-29.mp3	2026-09-08 05:31:42 Coordinated Universal Time	14.0 MB	Not analyzed
☐	marina_2026-09-08_04-50-41.mp3	2026-09-08 05:02:23 Coordinated Universal Time	16.1 MB	Not analyzed

Recordings may be played either by clicking on the desired link in the recording gallery, or by pressing the appropriate Play button. When listening to a recording, the time interval may be skipped forward either by advancing the slider with the mouse, or by clicking on an area of the waterfall. Clicking on the “**Digital white-noise Reduction**” checkbox will enable a simple white noise filter.

CW Decoding of recorded pass audio

Recordings
Folder: C:\bqe\bqe_wisp\plugins\bqe_sound_recorder\recordings

Buttons: Decode Checked Morse, Sort by Filename, Sort by Timestamp, Refresh

Choose a recording below.

☒ Digital white-noise reduction On — adaptive speech/CW filtering

☒ Play	marmotsat_2026-09-04_14-38-54.mp3	2026-09-04 14:51:21 Coordinated Universal Time	17.1 MB	▼ CQ DE VA7UVS CO-128 QRO 2 WATTS ANT Estimated 13.6 WPM; tone 375-656 Hz; 1251 pulses; 89% signal/timing confidence. Repeated message candidate (3 occurrences): CQ DE VA7UVS CO-128 QRO 2 WATTS ANT M DE VA7VH GOTV128 QRL I ATT DIPON IE AA.MARMOTSAT.CA CQ DE VA7UVS CO-128 QRO 2 WATTS ANT DIPOLE I ?MARMOTSAT.CA CQ D VA7UVS CO-128 QRO 2 WATTS ANT DIPOLE MWM.MARMOTSAT.CA CQ DE VA7UVS CO-128 QRO 2 WATTS ANT DIPOLE MWM.MARM ET+ECA CQ DE VA7UVS CO-128 QRO 2 WATTS ANT N MWM.LIT EMOTSAT.CA IUVS CO-128 QRO T.MARMOTSA EE S CO-128 QRN TARMOTSAT.CA OOTPE?EN QRO 2 WATTS ANT DI? E E TOTSAT.CI CQ DE VA7UVS CO-128 E ERO 2 WATTS ANT DIPOLE MWM.MARMOTSAT.CA CQ DE VA7UVS CO-128 QRO 2 WATTS ANT DIPOLE MWM.MARMOTSAT.D
☒ Play	cute-1_2026-09-04_12-10-49.mp3	2026-09-04 12:25:16 Coordinated Universal Time	19.9 MB	Too few regularly keyed pulses were found to decode Morse.

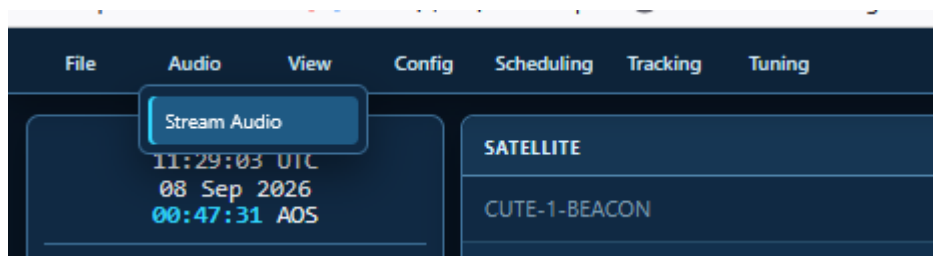
Recordings may be scanned for CW and decoded. If a message is repeated, the decoder will note how many copies of the same message were received, which can be used to judge accuracy of the transcription.

To use the decoder, select the checkbox to the left of the pass in question, or select the checkbox at the top of the recordings list, and click the “Decode Checked Morse” button at the top of the page.

The decoder is relatively tolerant of Doppler shift, however it may briefly drop a character when a frequency change happens. There is a balance between tuning too infrequently such that the audio slides out of the passband and adjusting tuning so often that the tuning itself causes dropouts. USB mode is recommended for situations with high doppler shift, as it allows for a greater drop in pitch before sliding out of the passband. Note that the CW decoder works via detecting and measuring

“pulses” in the audio. This means that static crashes, brief dropouts during tuning, or a radio that is squelched right at the background noise level may confuse the decoder and cause false decodes. Note that CW Decodes are not saved anywhere, but can be re-decoded at will by re-running the **“Decode Checked Morse”** button.

Audio Streaming



Audio from the radio sound-card can be streamed to the browser at any point via the Audio => Stream Audio menu selection. When streaming is enabled, the menu choice will switch to say “Disable Streaming”. Audio Streaming can be enabled at will, both during passes and during idle times, and will not interfere with any pass recording which has been configured. There is no provision to save streamed sound.

Tips and Release Notes

- Currently bqe_wisp has only been tested with the Yaesu FT-736r and FT-991a. Other rigs supported by Hamlib will likely work, providing that they cover the desired frequency ranges. For FT-736r users who have an extra band module, adding an intermediate frequency in bqe_config/my_rig.yaml will allow bqe-wisp to switch between mode b and mode J satellites, in a manner analagous to the “rev” button. Without a third band module, FT-736r users must be careful to not have U/V and V/U satellites in the same schedule.
- When using bqe_wisp with the FT-736r, the “Sat Mode” switch must be off. (The software will enable satellite_mode automatically.)
- For security reasons, yaml settings can be viewed via any browser session, however, saving changes can only be done via browser session running on the same computer as the server.
- When using bqe_wisp on a system where ‘python’ is not available, but python3 is, please either create an alias between python and python3, or update the python_path setting in bqe_config/general_settings.yaml, as it is needed for some of the scripts to call others during program operation.

```
- common:
  python_path: python # Can be a full path
```

- Currently all satellites need an uplink_frequency defined in satellites.yaml for the program to work correctly. In cases of receive only satellites (umka-1 and other sstv satellites) where no uplink exists, simply inventing one in a different band is fine. For the FT-736r, the uplink needs to be in a different band than the downlink. (This will be fixed in a later version.)
- All satellites need a ctcss tone defined in their satellites.yaml entry, regardless of whether the downlink is FM or some other mode. (This will be fixed in a later version.)
- Same band/frequency satellites (Such as the APRS digipeater) will not work with an FT-736r, due to a hardware limitation in the radio, but should work fine with an FT-991A or similar more modern radio.
- Presets are available and may be selected any time an idle task is running, however, idle tasks which run a program such as WSJTX which communicates directly via hamlib will block BQE Wisp from communicating with the radio, and thus will seem to ignore preset changes due to the inability to communicate with the radio. Stopping the idle task program manually before selecting a preset will work around this. In the specific case where wsjtx is desired, it may be possible to set up a wsjtx configuration which contains no radio configuration, and thus does not open its own channel to Hamlib.
- FDT is completely experimental, but I have been able to listen to well controlled stations for an entire RS-44 pass without needing to make any updates on my end.

Command Line Operation (For advanced users only)

BQE Wisp consists of 4 major python scripts intended to be executed by the user as well as several yaml configuration files, which will be explained later in this document.

Script	Arguments	Notes
bqe_update_keps.py	None	Downloads keps from several distribution sites to a file called keps.txt THIS CAN ALSO BE DONE FROM WITHIN THE UI.
bqe_schedule_passes.py	--nickname <nickname1> --	In cases where passes intersect,

nickname <another_nickname>
etc.

the pass with the highest maximum elevation will be scheduled. (Note – arguments are preceded by TWO dashes (minus signs.) --

THIS CAN ALSO BE DONE FROM WITHIN THE UI, See the Auto-Schedule section below. The web console is viewable at localhost:8028

bqe_wisp.py & bqe_wisp_web.py None

An additional script called **bqe_track_continuously.py** is called by **bqe_wisp** at pass time to perform tracking and tuning for the duration of the pass. This script can also be executed manually by itself via command line containing the necessary arguments, but there should be no need to do so. (Python **bqe_track_continuously.py** <nickname> (i.e. **bqe_track_continuously.py** iss)

Getting Started in Dos or Linux Shell Command line mode

- Either clone the bqe_wisp repo from github, or download a zip from wb1bqe@gmail.com google drive.
- Edit **bqe_config/my_qth.yaml** to contain your Callsign, Latitude / Longitude etc, as well as an idle task if required. (See the existing data for an actual example.)
- Edit **bqe_config/my_rig.yaml** to select the proper radio type and port etc. (Note: Radio type “1” is a dummy test, and can be used to try the software without sending any radio commands.)
- Edit **satellites.yaml** to add or update entries for any satellites you intend to use that are not already listed. This file is what ties a satellite name and catalog number to a nickname, and allows you to specify any extra information that the program may need. Fields are self explanatory (I hope) For now, Antenna Tracking is disabled, as the code refers to my own home brew antenna switch, and likely not of wide interest. The next release will support hamlib based rotators.
- After the initial configuration described above, to run the the program, simply open a command window and cd into the working directory and type the following:
- (run) **python bqe_update_keps.py**
- **python bqe_schedule_passes.py --nickname iss --nickname umka-1** etc. Note that the arguments are Nicknames which must have corresponding entries in “satellies.yaml”. If the same set of satellites will often be scheduled, the **--auto_schedule** flag may be used to schedule many satellites at once. **Auto_schedule** will attempt to schedule all satellites that have the

auto_schedule: True flag set in `bqe_config/satellites.yaml` Downloading of keps, and auto-scheduling can also be performed from within the UI, once the program has been initially configured and started. (A restart is required if the QTH has changed.)

```
nickname: "umka-1"
satellite_name: "UmKA 1 (RS40S)"
satellite_type: "SSTV"
catalog_number: '57172'
satellite_mode_required: True
downlink_frequency_mhz: '437.625'
downlink_mode: 'FM'
uplink_frequency_mhz: '145.850'
uplink_mode: 'FM'
bandwidth: '16000'
ctcss_tone: '0'
enable_antenna_tracking: False
enable_tuning: True
priority: '3'
auto_schedule: True
program_to_run_during_pass: "C:/Ham/MMSSTV/MMSSTV.EXE"
```

- (run) **python bqe_wisp.py**

Once started, **bqe_wisp.py** will load the schedule data created in the previous steps, and run through the schedule until all passes have been executed. For each pass, **bqe_wisp.py** will launch **bqe_track_continuously.py** to perform the tracking and tuning actions. Progress info is output to the command window, however, a more detailed UI is available via a browser window at <http://localhost:8028>.

Support

Best effort support is available via wb1bqe@gmail.com. Suggestions/feedback are also quite welcome!

After any kinks are worked out, my intention is to release this via Github as a public source offering, at which time, contributions via pull requests will also be welcomed.